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Amino acids are used to make proteins. **Table 1** shows the R groups of six different amino acids.

Table 1

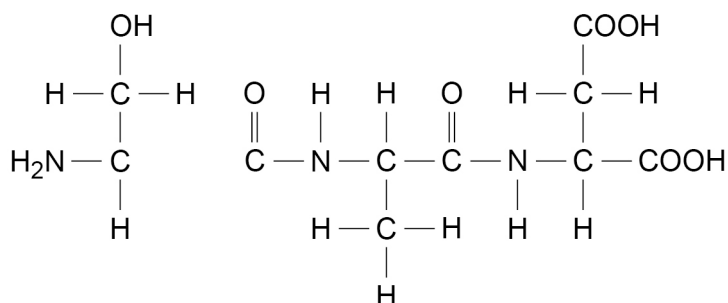
Amino acid	R group	Amino acid	R group
Alanine	CH ₃	Glutamic acid	CH ₂ CH ₂ COOH
Asparagine	CH ₂ CONH ₂	Glycine	H
Aspartic acid	CH ₂ COOH	Serine	CH ₂ OH

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Use **Table 1** to identify the **three** different amino acids used to make the polypeptide shown in **Figure 1**.

[2 marks]

Figure 1



Left amino acid _____

Middle amino acid _____

Right amino acid _____

Question 1 continues on the next page

Turn over ►

0 1 2

Table 2 shows three statements and names of four biological molecules.

Put a tick (✓) in each box where the statement is true for the biological molecule.

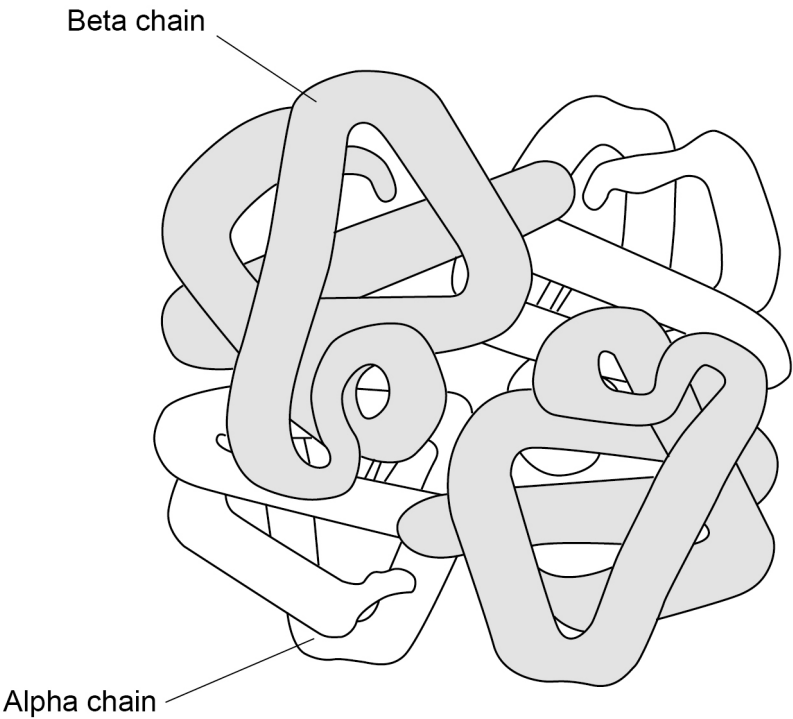
[3 marks]

Table 2

Statement	DNA	ATP	Reverse transcriptase	Phospholipid
Contains peptide bonds				
Is formed using a condensation reaction				
Is a polymer				

Figure 2 represents the structure of adult human haemoglobin.

Figure 2



0 1 3

The number of amino acids in the beta chains in **Figure 2** is 3.546% greater than in the alpha chains. Each alpha chain contains 141 amino acids.

Calculate how many amino acids there are in total in the haemoglobin molecule shown in **Figure 2**. Give your answer to the nearest whole number.

[2 marks]

Answer _____ amino acids

When a substance called BPG binds to haemoglobin, it reduces the affinity of haemoglobin for oxygen.

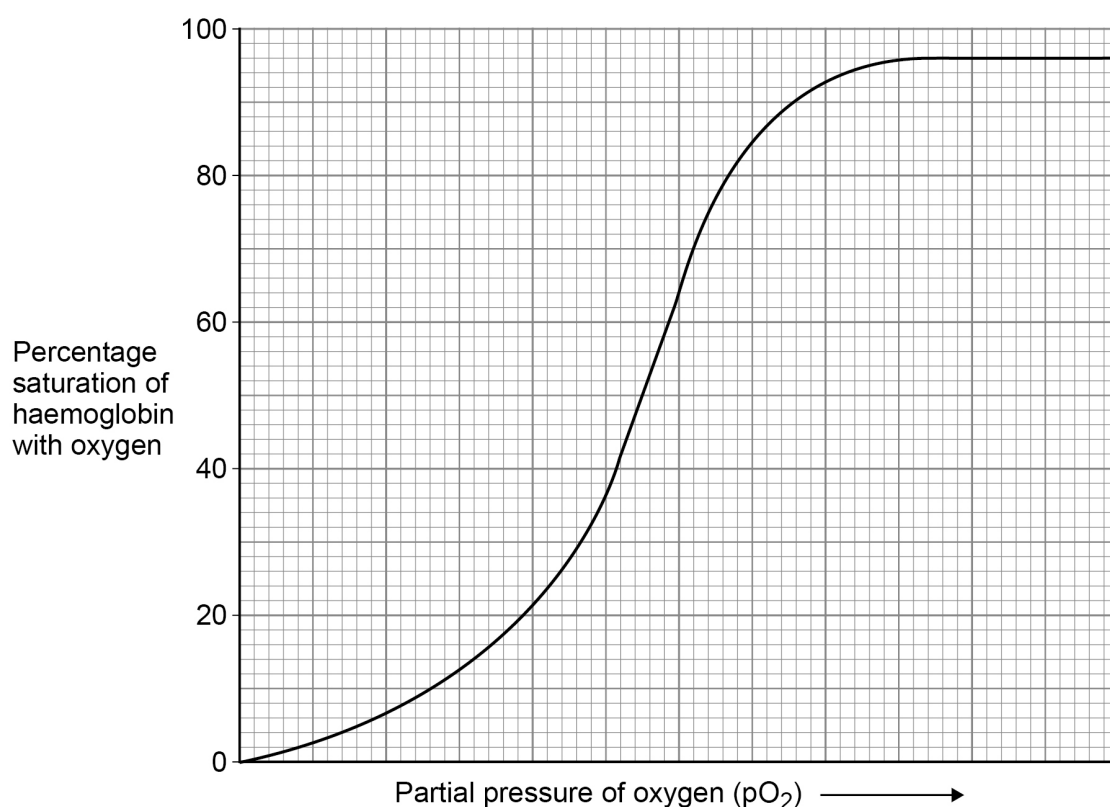
0 1 4

Figure 3 shows an oxyhaemoglobin dissociation curve for haemoglobin in normal conditions.

Sketch a curve on **Figure 3** to show the oxyhaemoglobin dissociation curve for haemoglobin when BPG binds to it.

[1 mark]

Figure 3



Question 1 continues on the next page

Turn over ►

0 1 . 5

Suggest and explain when it would be an advantage to a human for BPG to bind to haemoglobin.

[2 marks]

1

